The Classification of Supernovae Lightcurves with the Transient Name Server using Machine Learning and Neural Network Classification Algorithms

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Science Question | Motivations:

We seek to answer how well machine learning and neural networks can be implemented to the classification of supernovae lightcurves without the input of spectral identifiers. This is an interesting question because follow up observations are difficult to obtain and therefore spectra of these transient events are few and far between; however lightcurves are abundant. The problem with lightcurves is that they are not continuous and are generally very fragmented, so they require processing and curve fitting prior to introducing them to the machine learning algorithm.

Data:

We will be utilizing the Transient Name Server (TNS: https://www.wis-tns.org/) to identify already classified supernovae and join this table to a broker's table, such as Lasair (https://lasair.lsst.ac.uk) or ALeRCE (https://science.alerce.online/), in order to pull their corresponding lightcurves. From TNS we will pull lightcurves for 1000 SNIa, 1000 SNII, 416 SNIc, and 328 SNIb; from each of these sets, we will reserve 50 for use in model validation.

Methods:

We will first identify supernovae from TNS and will use this to identify lightcurves from a table of lightcurves. Then we will trim these lightcurves to the supernovae regime using the highest data point and carve around it by approximately 5 days before and 100 days after. We will then use a fitting method, such as Markov Chain Monte Carlo (MCMC), to fit a curve to the individual points. Some of these curves will be fed to a machine learning algorithm (ML) as training data. The rest of these curves will be then used to validate the machine learning model. Following this we will now pull not-yet-classified supernova lightcurves and apply the same trimming and fitting as above and then classify them with the ML. We will jointly be training a neural network using untrimmed, unfit lightcurves as well as validating it and feeding it not-yet-classified supernova lightcurves to compare its results to the ML.

Expected Results:

We are seeking to reject or not reject the use of lightcurves for identification of supernovae through the use of machine learning and neural networks. We will also compare the accuracy of, and report successful classification fractions from, the machine learning algorithms *and* the neural networks to determine the accuracy of the methods when compared to each other. This can consider the time required to train the neural network and the scope of data required to do so.